Store Sales Forecasting

Technical Project Report & Architecture Overview

Prepared for: Project Stakeholders

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1. Executive Summary

This technical report outlines the objectives, architecture, and technology stack for the Store Sales Forecasting project. The primary goal of this system is to build a robust and highly scalable machine learning pipeline capable of accurately predicting future sales across various store items, item families, and retail locations. By leveraging modern MLOps practices, the project ensures continuous integration, automated forecasting, and high-availability endpoints.

2. Project Objectives

Core Business & Technical Goals

The implementation of this project aims to achieve several critical business and operational goals:

- **Demand Forecasting:** Predict future sales over a specific horizon based on historical transaction data, enabling optimized inventory management and reducing stockouts or overstock scenarios.
- **Pattern Recognition:** Automatically identify complex seasonality patterns, long-term trends, and the measurable impact of external factors such as local holidays and promotional campaigns.
- **Automated Machine Learning Pipeline:** Implement an end-to-end operational ML pipeline that encompasses automated data ingestion, advanced preprocessing, model training, and continuous prediction generation.
- **Scalable Model Deployment:** Serve the trained machine learning models via a highly scalable, low-latency RESTful API, facilitating seamless integration with internal dashboards or frontend applications.
- **Reproducibility & Governance:** Ensure that all model experiments, datasets, and hyperparameters are meticulously tracked and versioned using state-of-the-art MLOps tooling.

3. Technology Stack & Architecture

The project architecture leverages a modern, Python-centric Data Science and Machine Learning Operations (MLOps) stack, specifically selected for handling high-volume time-series data.

Machine Learning & Data Processing

- **Python 3:** The core programming language orchestrating data pipelines and API services.
- **Pandas & NumPy:** Essential libraries used for high-performance data manipulation, cleaning, and sophisticated feature engineering.

- **LightGBM & Scikit-learn:** High-performance, distributed gradient boosting frameworks utilized for training the predictive regression models.
- **Darts:** A specialized Python library that provides user-friendly and advanced forecasting algorithms explicitly designed for multivariate time-series.
- **Hampel:** A robust filtering algorithm used for reliable outlier detection and removal in volatile time-series datasets.

API Infrastructure & Deployment

- **FastAPI:** A modern, high-performance web framework used for building the prediction REST APIs.
- **Uvicorn:** A lightning-fast ASGI web server implementation serving the FastAPI endpoints.
- **Docker:** Containerization technology ensuring absolute consistency across local, staging, and production environments.

MLOps & Data Versioning

- **DVC (Data Version Control):** An essential tool utilized to version massive datasets and strictly track machine learning models and experiments.
- **Joblib:** Employed for serializing and efficiently saving/loading the trained LightGBM model artifacts.

Database & Storage

- **PyMongo:** The MongoDB driver for Python, architected to store raw sales records, system telemetry, application logs, and historical prediction data.

4. Conclusion & Next Steps

The chosen technology stack and architectural design present a highly capable foundation for enterprise-scale time-series forecasting. The combination of LightGBM and Darts offers unparalleled accuracy for sales predictions, while FastAPI and Docker guarantee that the system can scale to meet real-time production demands. The integration of DVC ensures that the project remains maintainable, reproducible, and ready for future iterations.